AI.301

Virtual Reality

COURSE OVERVIEW

Immersive synthetic environments: displays, tracking, interaction, and VR for training and simulation.

VIRTUAL REALITY

1

SESSION

60 MINUTES

ORIENTATION: STEPPING INSIDE

KEY TOPICS

- What is VR
- Immersion vs. presence
- VR hardware: HMDs, controllers
- VR in training and robotics

TEACHING FOCUS

Define virtual presence.

KEY TAKEAWAY VR replaces the physical with the synthetic.

2

SESSION

60 MINUTES

VR HARDWARE & DISPLAYS

KEY TOPICS

- Head-mounted displays
- Lenses and distortion
- Tracking systems
- Input devices

TEACHING FOCUS

Build the sensory interface.

KEY TAKEAWAY Presence begins with the display.

VIRTUAL REALITY

3

SESSION

60 MINUTES

3D INTERACTION

KEY TOPICS

- Selection and manipulation
- Wayfinding and navigation
- Gestures in VR
- Haptic feedback

TEACHING FOCUS

Let users act inside virtual worlds.

KEY TAKEAWAY Interaction makes VR participatory.

4

SESSION

60 MINUTES

RENDERING FOR VR

KEY TOPICS

- Stereo rendering
- Latency and frame rate
- Distortion correction
- Performance budgets

TEACHING FOCUS

Render two views fast enough.

KEY TAKEAWAY VR demands low latency and high frame rates.

VIRTUAL REALITY

5

SESSION

60 MINUTES

VR FOR ROBOTICS

KEY TOPICS

- Robot telepresence
- VR training environments
- Sim-to-real VR
- Human-in-the-loop simulation

TEACHING FOCUS

Train and operate robots in VR.

KEY TAKEAWAY VR is a safe training ground for robots.

VIRTUAL REALITY

6

SESSION

60 MINUTES

VR SYSTEMS & REVIEW

KEY TOPICS

- VR prototype demo
- Course recap and assessment
- Toward CPS (AI.302)
- User study

TEACHING FOCUS

Build and evaluate a VR experience.

▲ **KEY TAKEAWAY** Presence is measured by the user.